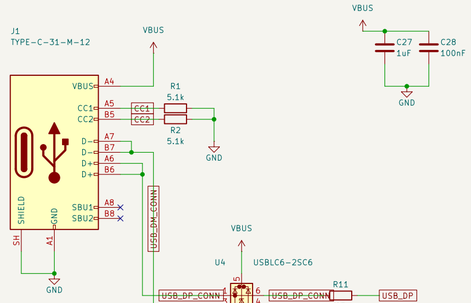
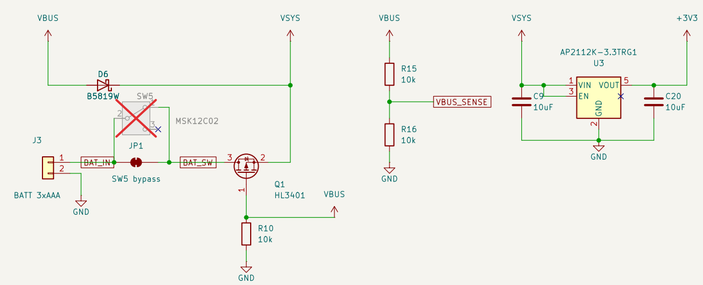
INTENT 2026 hardware badge — generated by 03-kicad/scripts/gen_sch.py from design.py (edit design.py, not this file).

USB-C (USB 2.0 full speed, device)



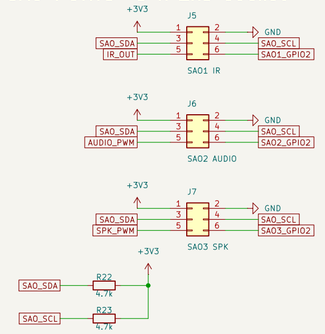
HRO TYPE-C-31-M-12 (C165948). 5.1k on CC1/CC2 = UFP.
USBLC6-2 ESD array, 27R series (RP2040 guide).
C27/C28 VBUS decoupling. SBU pins unused.

POWER USB or 3xAAA -> VSYS -> 3V3 LDO



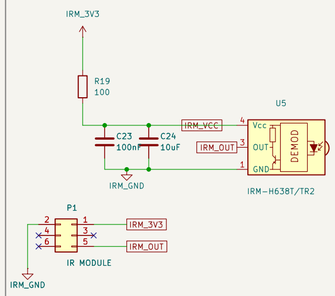
VSYS = 4.65 V on USB (after D6), 3.0-4.8 V on battery. With both present VBUS holds Q1 off, D6 blocks back-feed.
SW5 is DNP (the switch is on the battery box); JP1 bridges it. Cut JP1 only if SW5 is fitted.
RGB LEDs run from VSYS, everything else from 3V3 (AP2112K-3.3, 600 mA, EN tied to VIN).
R15/R16: VBUS_SENSE = 2.5 V on USB, 0 V on battery (GPIO29 / ADC3) so firmware can dim and sleep.

SAO PORTS 3 x 2x3 socket



SAO (badge.team): 1 3V3, 2 GND, 3 SDA, 4 SCL, 5 GPIO1, 6 GPIO2.
I2C0 on GPIO12/13, 4.7k pull-ups. GPIO1: IR 22 / AUDIO 11 / SPK 25.
GPIO2: GPIO26/27/28 (ADC). Socket on badge, 2x3 pin header on the add-on.

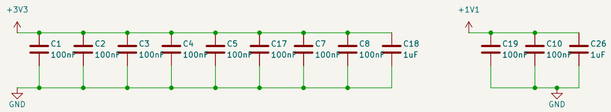
IR MODULE breakaway board, plugs into J5

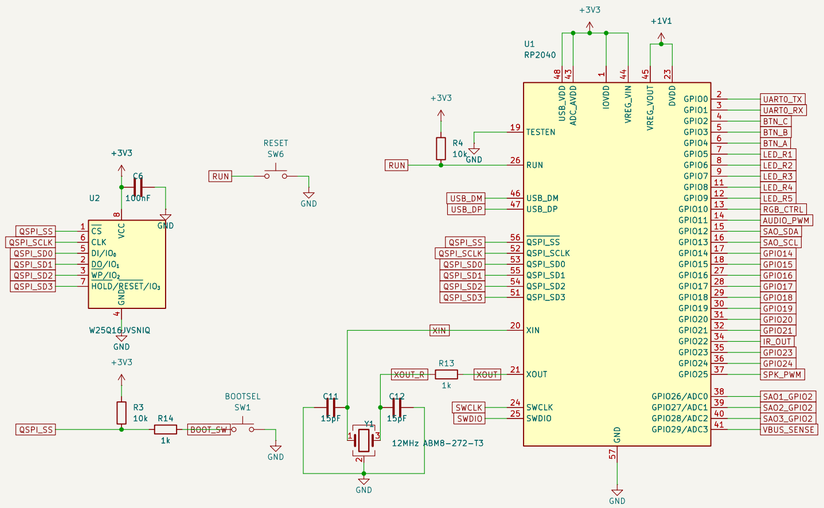


Everlight IRM-H638T, 38 kHz, Open-collector out, active low, internal pull-up.
R19 + C24 = datasheet RC filter. Pinout 1,2 = GND, 3 = OUT, 4 = VCC (verified).
P1: SAO 2x3 header, 1 = 3V3, 2 = GND, 5 = GPIO1 = OUT; plugs into J5. IRM_* nets live only on this board;
no copper crosses the tab. Snap off at the mouse bites, plug in standing up (lens looks sideways).

RP2040 dual Cortex-M0+, 2 MB QSPI flash, 12 MHz crystal

Decoupling (RP2040 guide 2.1.2/2.1.3): 100 nF per IOVDD (6x), USB_VDD, ADC_AVDD, DVDD (C19 pin 23, C10 pin 50): 1 uF on VREG_VIN (C18) and VREG_VOUT (C26).

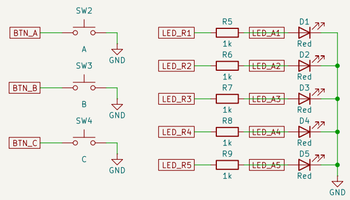




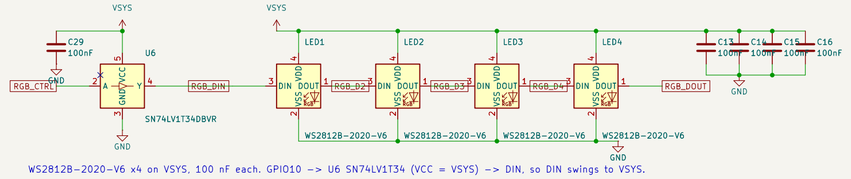
BOOTSEL: hold SW1 while plugging USB in -> UF2 bootloader. R14 protects the QSPI_SS driver.
Crystal: Abracon ABM8-272-T3 (CL 10 pF, ESR 50 ohm) with 15 pF loads + 1k, exactly the RP2040 guide 2.3 circuit.
TESTEN tied to GND. RUN: 10k pull-up, SW6 RESET to GND, also on J2. BOOTSEL + tap RESET = UF2 mode.

GPIO: 0/1 UART0 -> TP1/TP2; 4/3/2 = keys A/B/C;
5-9 red LEDs; 10 WS2812 data (via U6); 11/22/25 SAO GPIO1;
12/13 SAO I2C0; 26-28 SAO GPIO2 (ADC0-2); 29 VBUS sense;
14-21, 23/24 free on J2 (SPI1, UART1).

USER INTERFACE keys, credit LEDs, RGB shelf

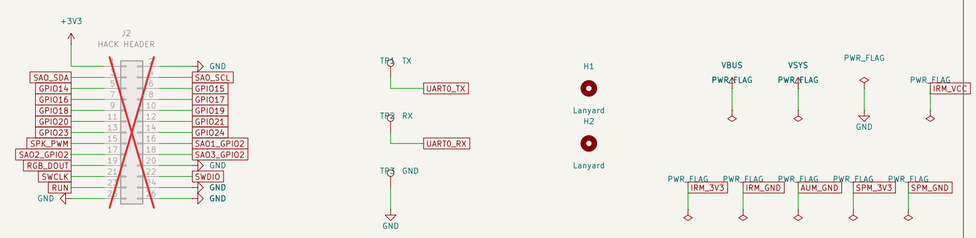


Keys A/B/C: active low, internal pull-ups, debounce in firmware.
Credit display: 5 red LEDs, 1k series -> ~1.5 mA each (dim by design).



WS2812B-2020-V6 x4 on VSYS, 100 nF each. GPIO10 -> U6 SN74LV1T34 (VCC = VSYS) -> DIN, so DIN swings to VSYS.
LED VDD spec is 3.7-5.3 V: firmware dims/stops the LEDs on low battery (VSYS < ~3.7 V). Chain out to J2 pin 19.

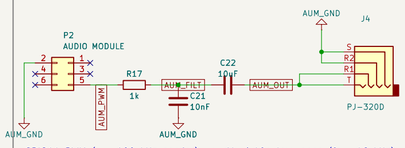
HACK HEADER J2 (DNP), test points, lanyard holes



2x13, 2.54 mm, not fitted — players solder their own pins. 3V3 logic only, no 5 V on the header.
Pin 1 = top-left square pad. GND on both end rows: a reversed ribbon shorts 3V3 to GND at worst (LDO limits).
10 free GPIO + SAO I2C/GPIO2 + SPK_PWM + SWD + RUN + RGB out. TP1-TP3: UART0 115200 8N1. H1/H2: 4.0 mm lanyard holes.

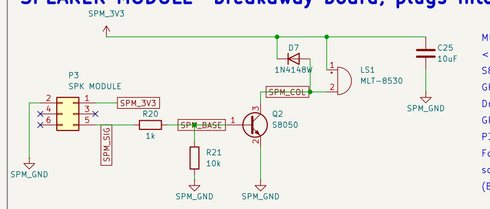
PWR_FLAGs: nets fed only through passive pins
(row 2: breakaway-module rails and grounds).

AUDIO MODULE breakaway board, plugs into J6



GPIO11 PWM (>= 100 kHz carrier) -> 1k / 10n low-pass (fc ~ 16 kHz) -> 10 uF DC block -> tip + ring 1 (mono). Sleeve + ring 2 = GND.
Level into 32 ohm ~ 0.1 Vpp. PJ-320D (C431535) SMD pads: 1 = S, 2 = R2, 3 = R1, 4 = T. Detect-switch pads unused.
P2: SAO 2x3 header, 2 = GND, 5 = GPIO1 = PWM (3V3 unused); plugs into J6. Snap off; the jack then points sideways.

SPEAKER MODULE breakaway board, plugs into J7



MLT-8530 magnetic transducer: 2.7 kHz resonance, 16 ohm,
<= 95 mA at 2.5-4.5 V, so it runs from the 3V3 LDO.
S8050 low-side switch, 1k base; 10k keeps it off while the
GPIO floats at reset. 1N4148W flyback, 10 uF local bulk.
Drive: 2.7 kHz square wave (50 % PWM) on GPIO25.
GPIO25 is also J2 pin 15 (shared).
P3: SAO 2x3, 1 = 3V3, 2 = GND, 5 = GPIO1 = SIG (J7).
Footprint wires the two dummy pads crosswise,
so a 180 deg placement only reverses the coil.
(BOM-B2 add-on 7.)

Generated by scripts/gen_sch.py — edit design.py, not this file
RP2040 badge, 56x100 mm vending-machine outline + 3 breakaway SAO modules (IR, audio, speaker)

INTENT 2026 — by Palo Alto Networks

Sheet: /
File: intent-2026-badge.kicad_sch

Title: INTENT 2026 Hardware Badge

Size: A2	Date: 2026-10-17	**Rev: 0.6**
KiCad E.D.A. 10.0.6		Id: 1/1